

Introduction to Machine Learning

Lecture 2

25 January 2022

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Outline

- Overview of the last class
- Basic Terminologies widely used in ML
- Types of Machine Learning



Introduction

- Machine Learning
 - Grew out of work in AI
 - New capability for computers
- ML has revolutionized the way we live in a bigger way
- Some Real time Examples which are part of our daily lives
 - Computer Vision (Cameras, Object detection, Recognition), Natural language processing, handwriting recognition, Self driving cars,



What is Machine Learning ?

Definition

Arthur Samuel (1959). Machine Learning:

Field of study that gives computers the ability to learn without being explicitly programmed.



Tom Mitchell (1998). Well-posed Learning Problem:

A computer program is said to *learn* from **experience E** with respect to some **task T** and some **performance measure P**, if its performance on T, as measured by P, improves with experience E.



Kevin Murphy (2012):

Machine learning is the study of computer algorithms that improve automatically through experience.



“A computer program is said to *learn* from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

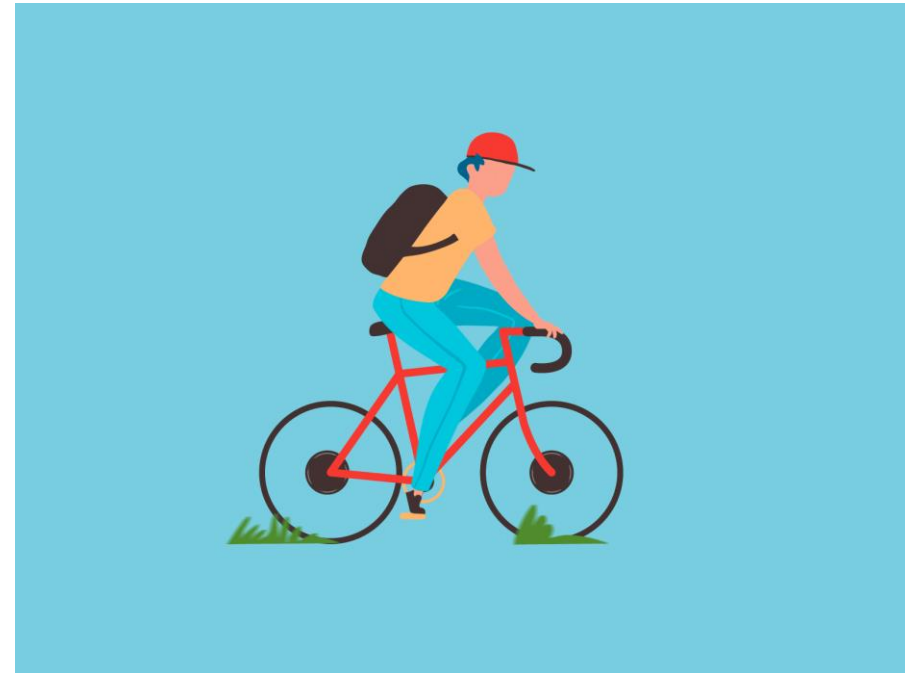
Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. Please categorize the following ?

- a. Classifying emails as spam or not spam.
- b. Watching you label emails as spam or not spam.
- c. The number (or fraction) of emails correctly classified as spam/not spam.

True Challenge to ML

- Is to solve the tasks that are easy for people to perform but hard for people to describe formally

Can some one tell the example ???



What ML Consists of ?

- Statistics, Optimization, Probability, Information Theory, **Psychology**, **Philosophy**, Neurobiology, Control Theory, Computer Science, Programming, Calculus, Linear Algebra, etc.



When Machine Learning is feasible ?

What we need/ or when can we apply ML ?

1. We need Data. {Without Data ML can not be applied}
Data can be: Numbers, Text, Image, Audio, Speech, etc.

What else we can say about the data ? Or What you expect from the data ?

2. The Data Should follow some pattern
Patterns means— It should be mathematically expressible



Pattern is there, what's next ?

Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...



Features and Label

Label- Ground truth/output

Features-Inputs/variables

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	yes
D6	Rain	cool	Normal	Strong	Yes
D7	Overcast	Cool	High	Weak	No
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Person playing tennis on a particular day based om weather conditions



Lets Check out with some other data sets

Lets see what is there in the Data Sets:

- House Price Prediction
- Iris Data Set
- Wisconsin Data set



Repositories to find the Data Sets

<https://www.kaggle.com/datasets>

<https://archive.ics.uci.edu/ml/index.php>



Machine Learning Terminologies

- Labelled Data
- Un-labelled Data
- Classification
- Regression
- Clustering
- Train Data
- Test Data
- Features

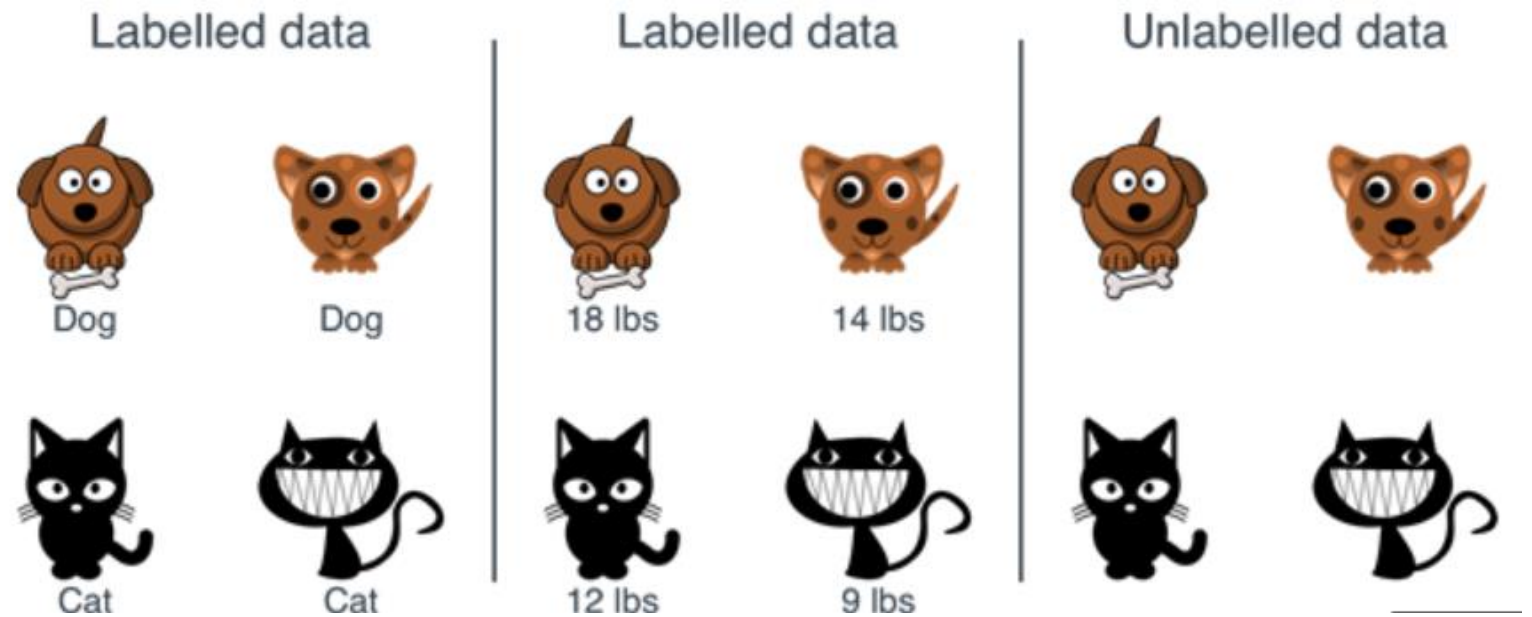


Labelled Data

Data Comes with Label

The label can be **name**, **number**, **Boolean** (Yes or No), etc.

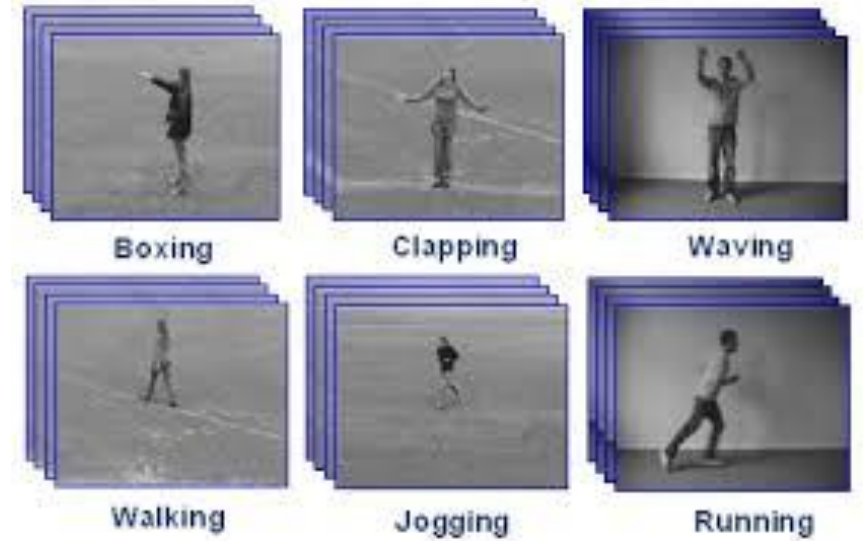
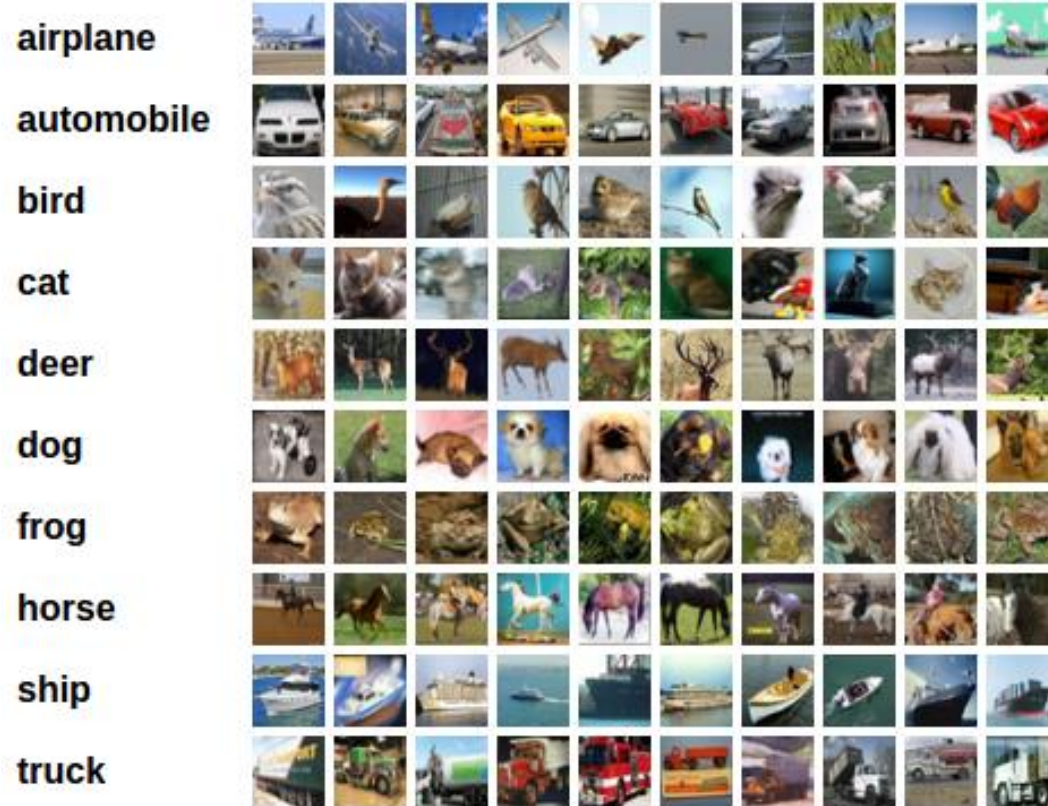
- **Labeled data** is a group of samples that have been tagged with one or more labels.
- Labeling typically takes a set of unlabeled data and augments each piece of it with informative tags.
- For example, a data label might indicate whether a photo contains a horse or a cow, which words were uttered in an audio recording, what type of action is being performed in a video



Source: Kaggle



Few Examples



Action recognition

How about Voice/Speech labelling ?



Unlabeled Data

Unlabeled data is a designation for pieces of data that have not been tagged with labels identifying characteristics, properties or classifications

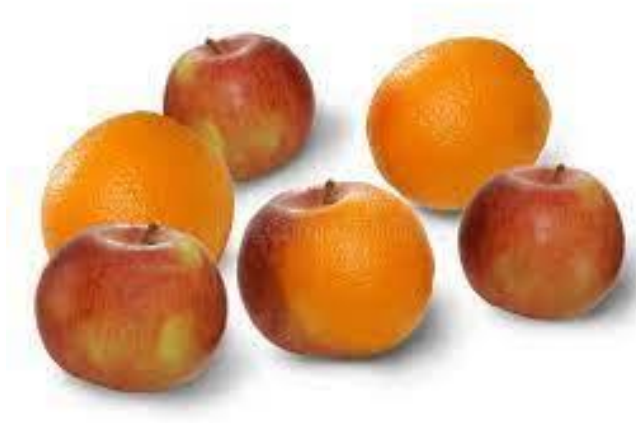
For example:

If I want to develop an automated algorithm for detecting the tumor in CT image.

If we approach the Hospital, they will give all the images without any labels, then we call it as unlabeled data.

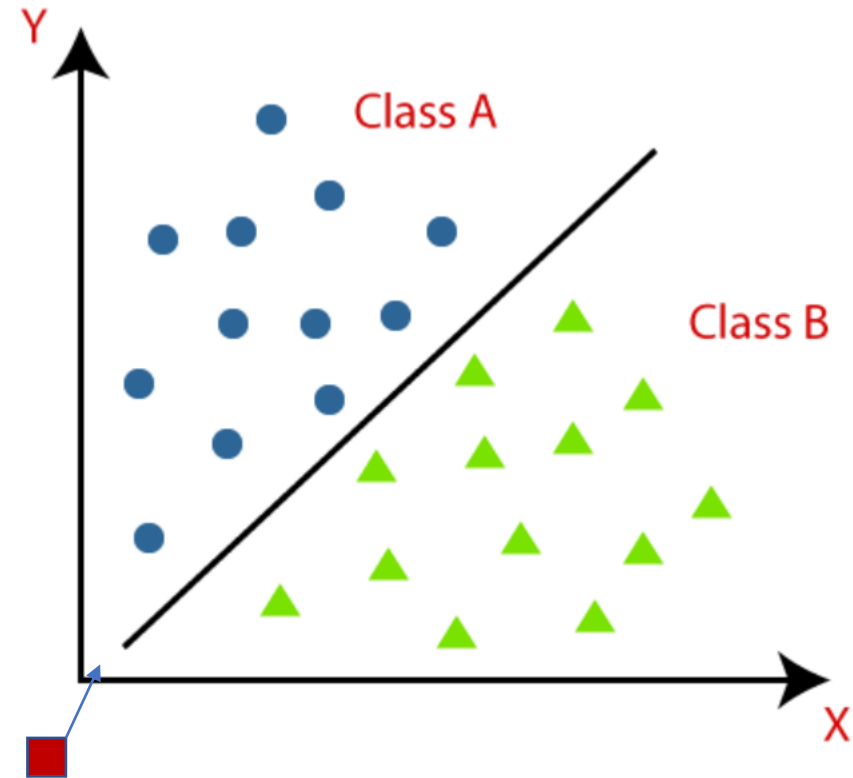
If the hospital, gave us the medical images along with its diagnosis report, then we call it as an labelled data.





Classification

- In Classification, a program learns from the given dataset or observations and then **classifies new observation into a number of classes or groups**. Such as, Yes or No, 0 or 1, Spam or Not Spam, cat or dog, etc.
- **Classes can be called as targets/labels or categories.**



Regression

- If algorithm wants to predict a continuous variable, then we call it as an Regression Problem.

Ex: Predicting the price of a house based on cost. Predicting how many items will be sold in a shop in a given day.

Which of the below falls under Regression problem: ?

- Predicting age of a person
- Predicting nationality of a person
- Predicting whether stock price of a company will increase tomorrow
- Predicting whether a document is related to sighting of UFOs?



Train Data, Test Data, Features

To evaluate the algorithm. Datasets, are divided into two categories

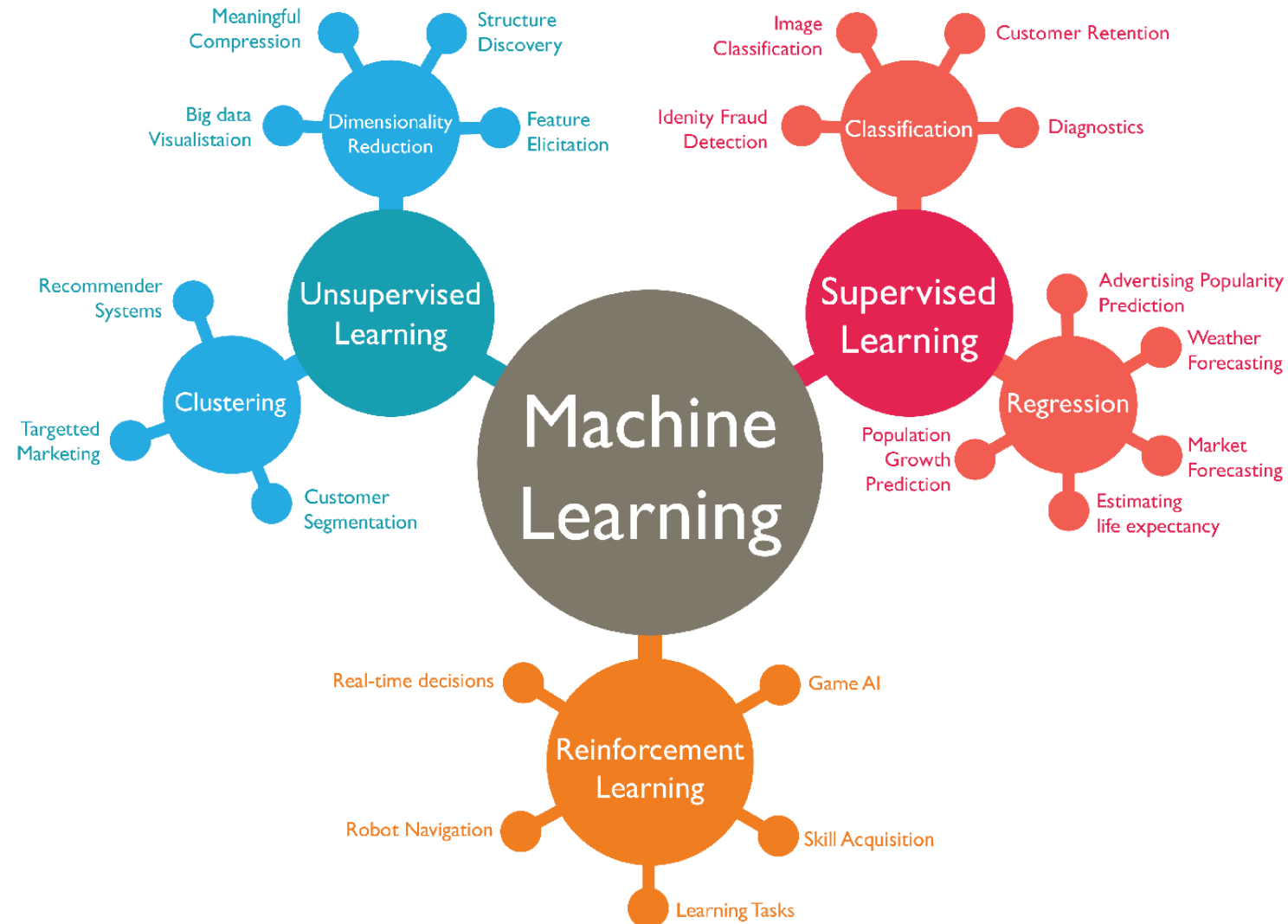
Training datasets and Test data sets

Algorithms are trained on Training Data, and it is evaluated on Test Data.

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	yes
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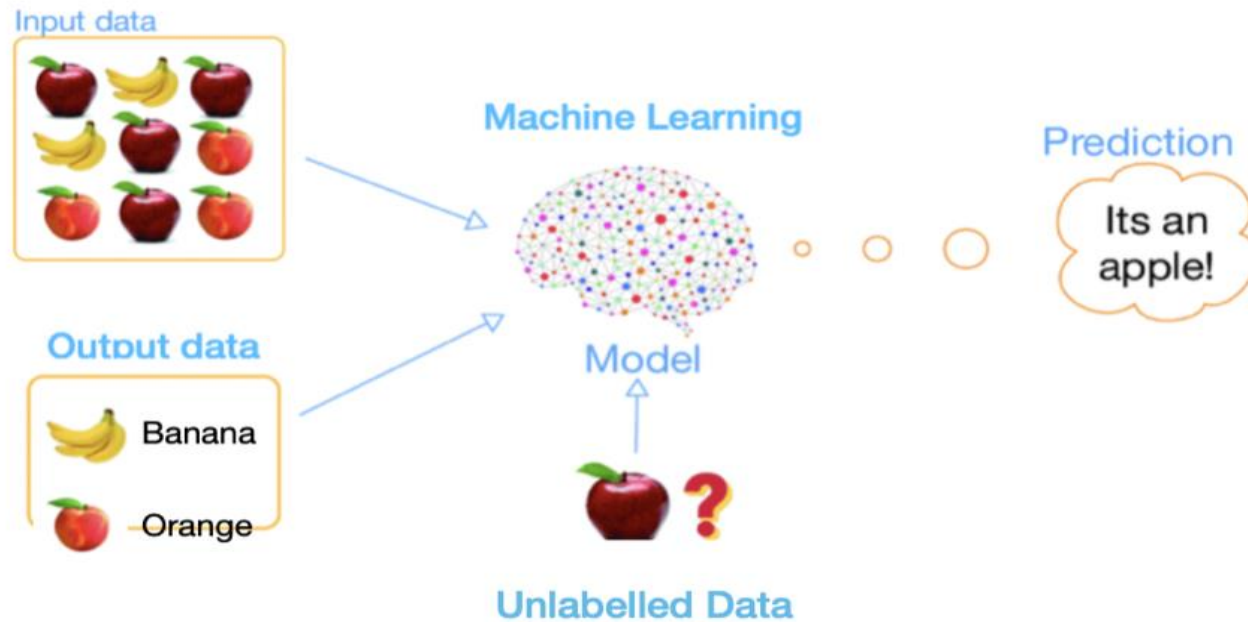


Different Types of Machine Learning

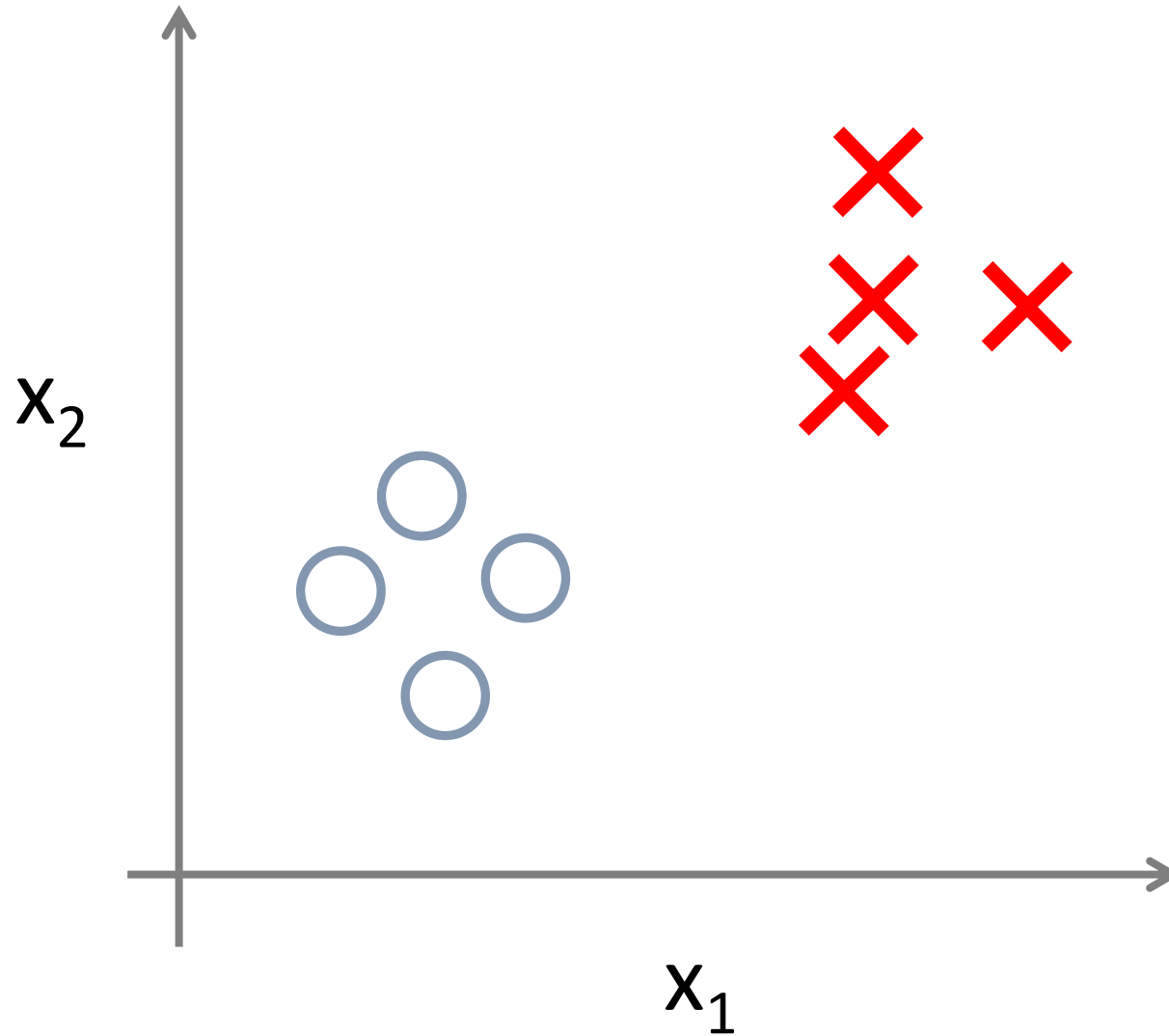


Supervised Learning

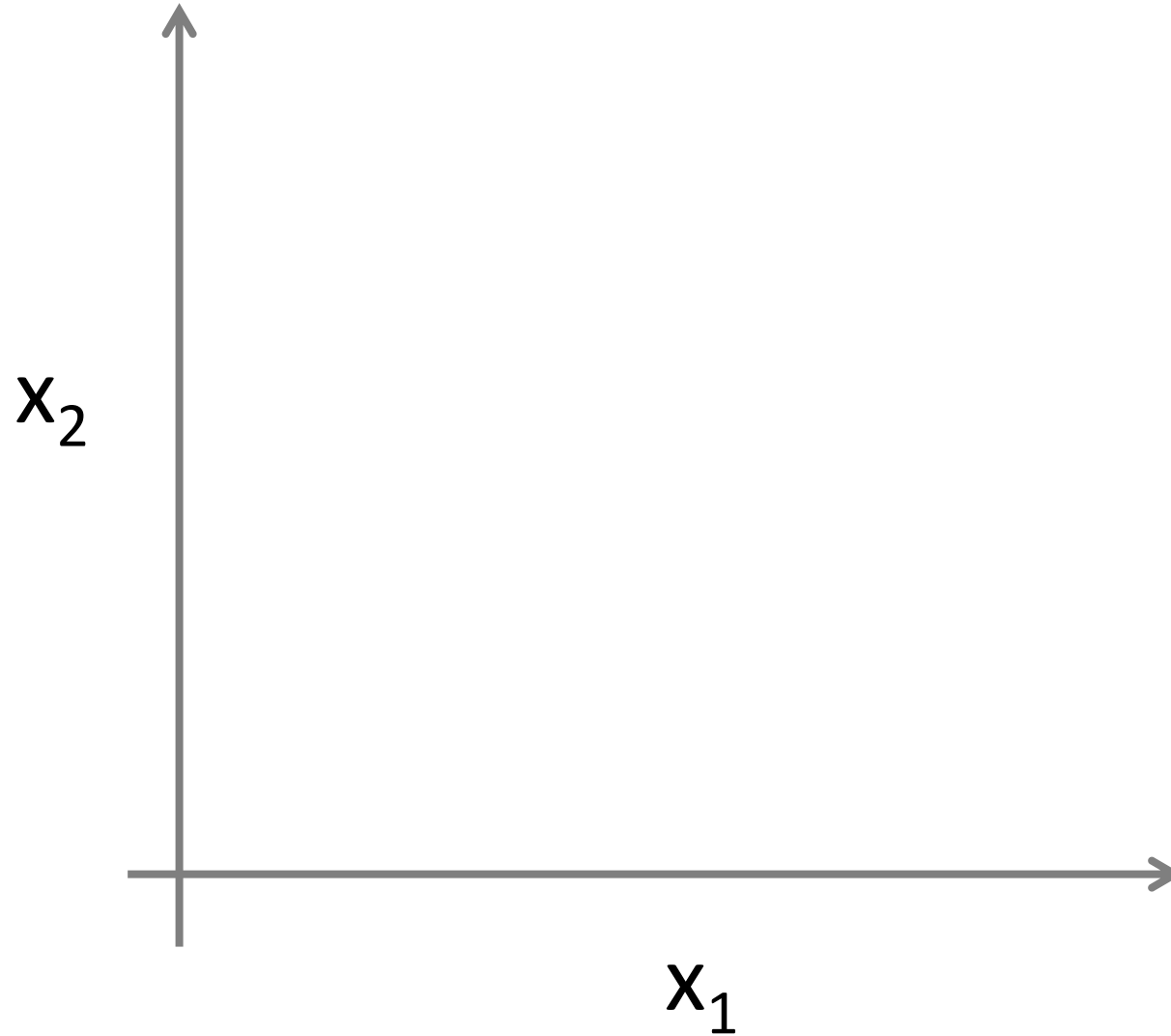
Supervised learning is the types of machine learning in which machines are trained using well "labelled" training data.



Supervised Learning- Classification



Supervised Learning- Regression



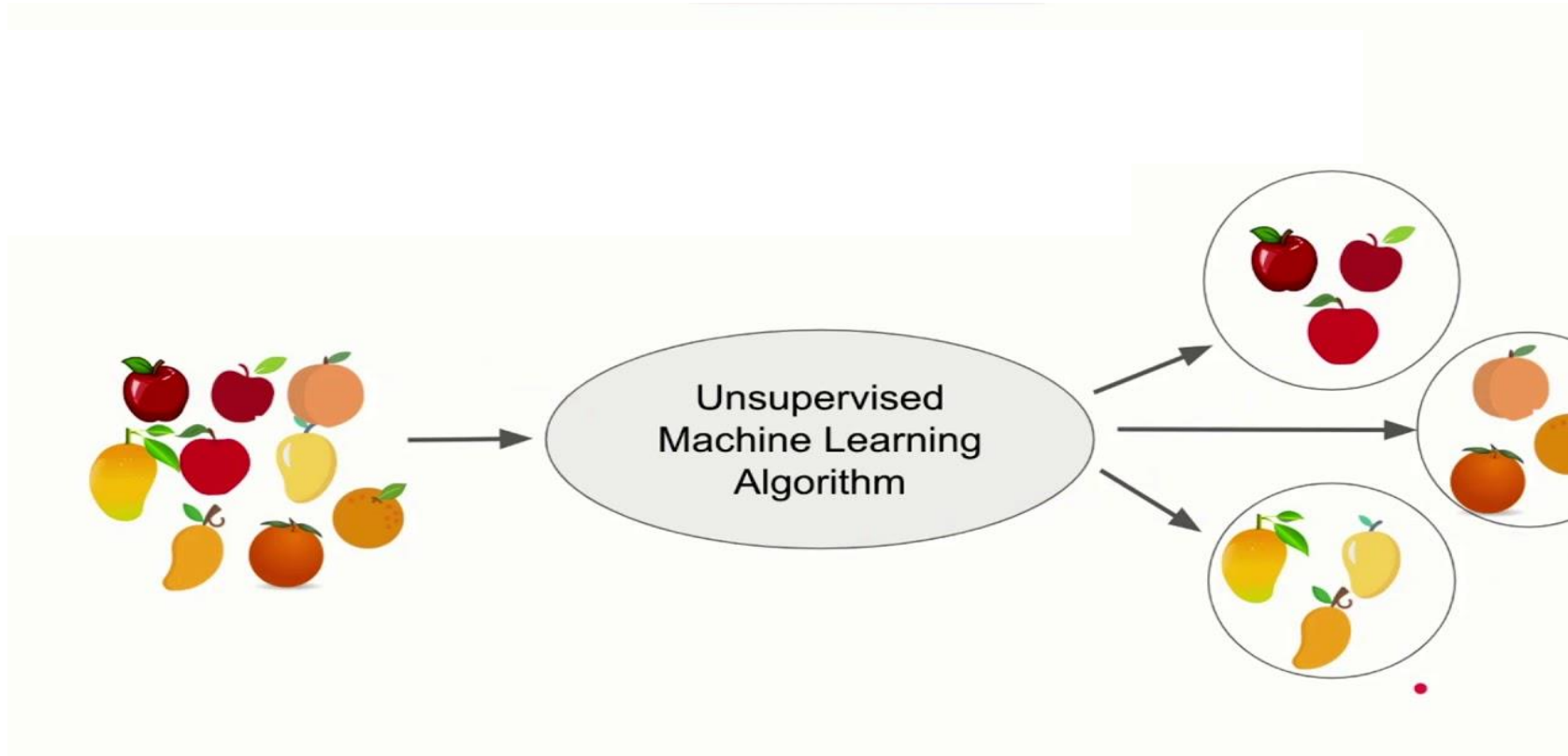
Regression Examples

- Predicting the Stock Price
- Predicting the Temperature
- Predicting the Sugar levels
- Predicting the Gold Price
- Predicting the Projected score in a cricket match

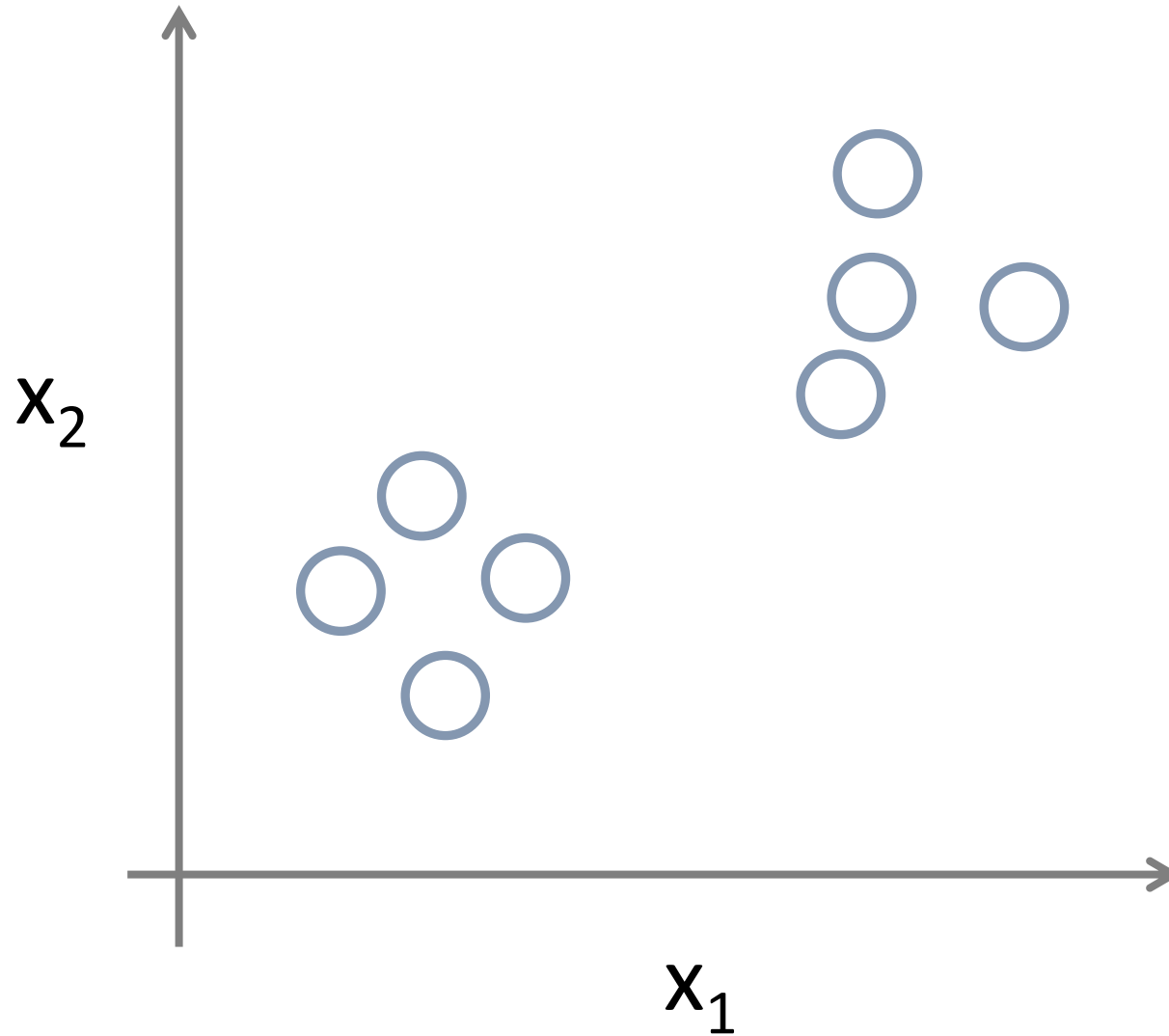


Unsupervised Learning

- Unsupervised learning is a type of [machine learning](#) in which the **algorithm is not provided with any pre-assigned labels or scores for the training data**. As a result, unsupervised learning algorithms must first self-discover any naturally occurring patterns in that training data set.



Unsupervised Learning



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Christine O'Donnell »

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GOP tea party backer defends Christine O'Donnell The Associated Press

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US Stocks Climb After Recession Called Over, Homebuilders Gain

MarketWatch - Kristina Peterson - 16 minutes ago

NEW YORK (MarketWatch) -- US stocks climbed Monday, gaining speed after a key nonprofit organization officially called the recession over, giving investors a boost of confidence in the gradual economic recovery.

Longest recession since 1930s ended in June 2009, group says

Los Angeles Times

Downturn Was Longest in Decades, Panel Confirms New York Times

Wall Street Journal - AFP - CNN - USA Today all 276 news articles »

Deepwater Horizon »

BP Oil Well, Site of National Catastrophe, Dies at One

Vanity Fair - Juli Weiner - 22 minutes ago

The BP oil well, site of the Deepwater Horizon explosion that led to the worst oil spill in US history, died today at one year old.

Video: Blow-out BP Well Finally Killed in Gulf

The Associated Press

Weiss Doubts BP Would End Operations in Gulf of Mexico: Video Bloomberg

CNN International - Wall Street Journal (blog) - The Guardian - New York Times all 2,292 news articles »

Recent

Recession officially ended in June 2009

CNNMoney - Chris Isidore - 39 minutes ago

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USA Today - Gerry Broome - 5 minutes ago

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Crisis response: Pakistan floods

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San Jose Mercury News - 48 minutes ago

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Spotlight

Sarkozy rages at EU 'humiliation'

Financial Times - Peggy Hollinger - Sep 16, 2010

BP Kills Macondo, But Its Legacy Lives On

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BP Kills Macondo, But Its Legacy Lives On

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By James Herron

BP confirmed late Sunday that the Macondo well that leaked almost five million barrels of oil into the Gulf of Mexico has been permanently sealed, but the well will continue to affect BP and the wider oil industry for many years.

The most immediate worry for BP and its shareholders is how the authorities will apportion blame for the spill. BP's own investigation spread responsibility across

Fire boat response crews battled the blazing remnants of the off shore oil

Associated Press

on Deepwater Horizon on April 21, 2010.

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Allen: Well is dead, but much Gulf Coast work remains

By the CNN Wire Staff

September 20, 2010 — Updated 1317 GMT (2117 HKT)

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What next for Gulf oil spill?

STORY HIGHLIGHTS

(CNN) — The ruptured Macondo well, a mile under the Gulf of Mexico off the Louisiana coast, has been pronounced dead

BP oil spill cost hits nearly \$10bn

www.guardian.co.uk/environment/2010/sep/20/bp-oil-spill-dee

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BP oil spill cost hits nearly \$10bn

BP has set up a \$20bn compensation fund after the Deepwater Horizon disaster, which has so far paid out 19,000 claims totalling more than \$240m

Julia Kollewe

guardian.co.uk, Monday 20 September 2010 08:33 BST

Article history

BP's costs for the Deepwater Horizon disaster have hit \$10bn. Photograph: Ho/Reuters

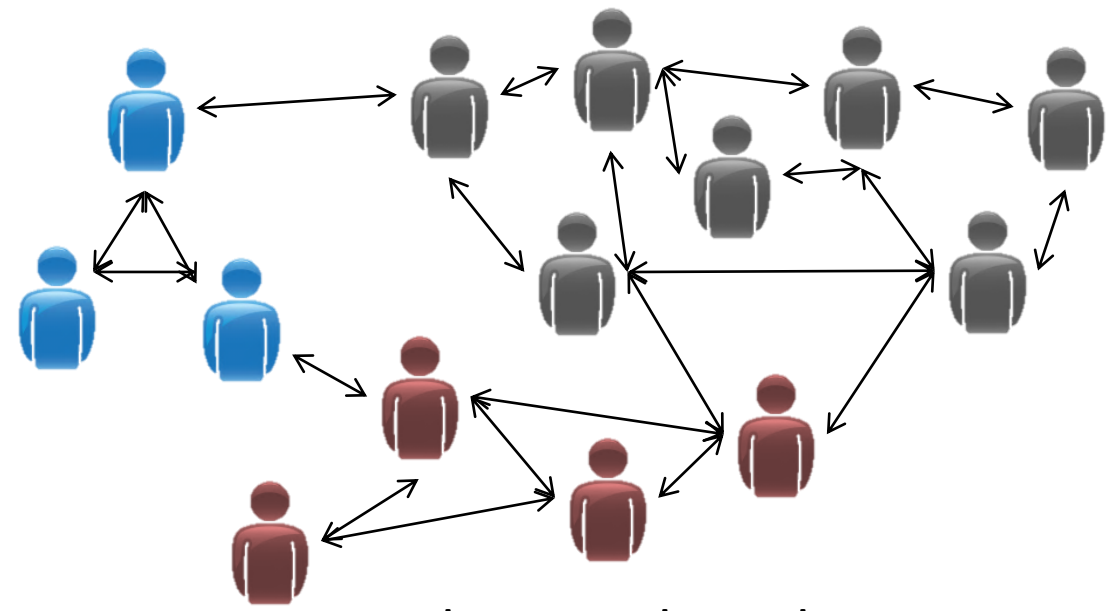
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Machine Learning

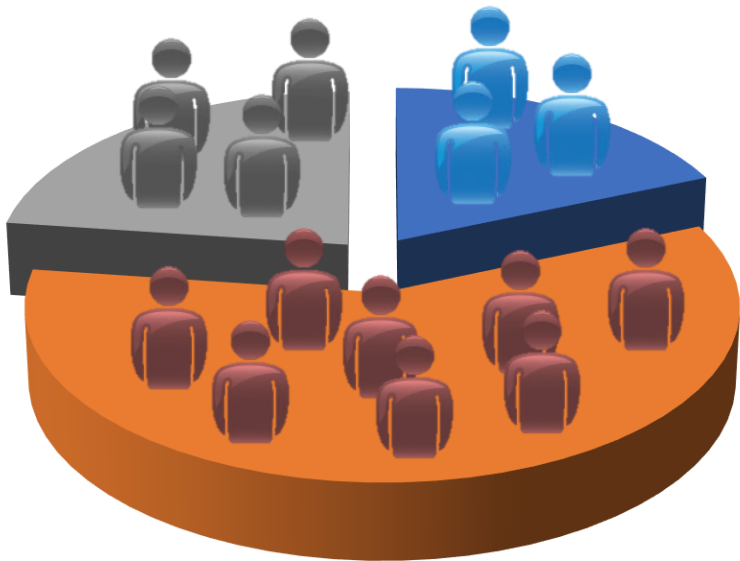




Organize computing clusters



Social network analysis



Market segmentation



Image credit: NASA/JPL-Caltech/E. Churchwell (Univ. of Wisconsin, Madison)

Astronomical data analysis



Reinforcement Learning

- Reinforcement learning is an area of machine learning concerned with how intelligent agents ought to take actions in an environment in order to maximize the notion of cumulative reward.
- Example: Gaming, Autonomous helicopter, Recommender Systems



What kind of learning algorithms will apply for below applications

1. Given email labeled as spam/not spam, learn a spam filter.
2. Given a set of news articles found on the web, group them into set of articles about the same story.
3. Given a database of customer data, automatically discover market segments and group customers into different market segments
4. Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

Time Table Slots: Machine Learning

Theory Hours:



Linear Regression

Training set of housing prices (Portland, OR)

Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

Notation:

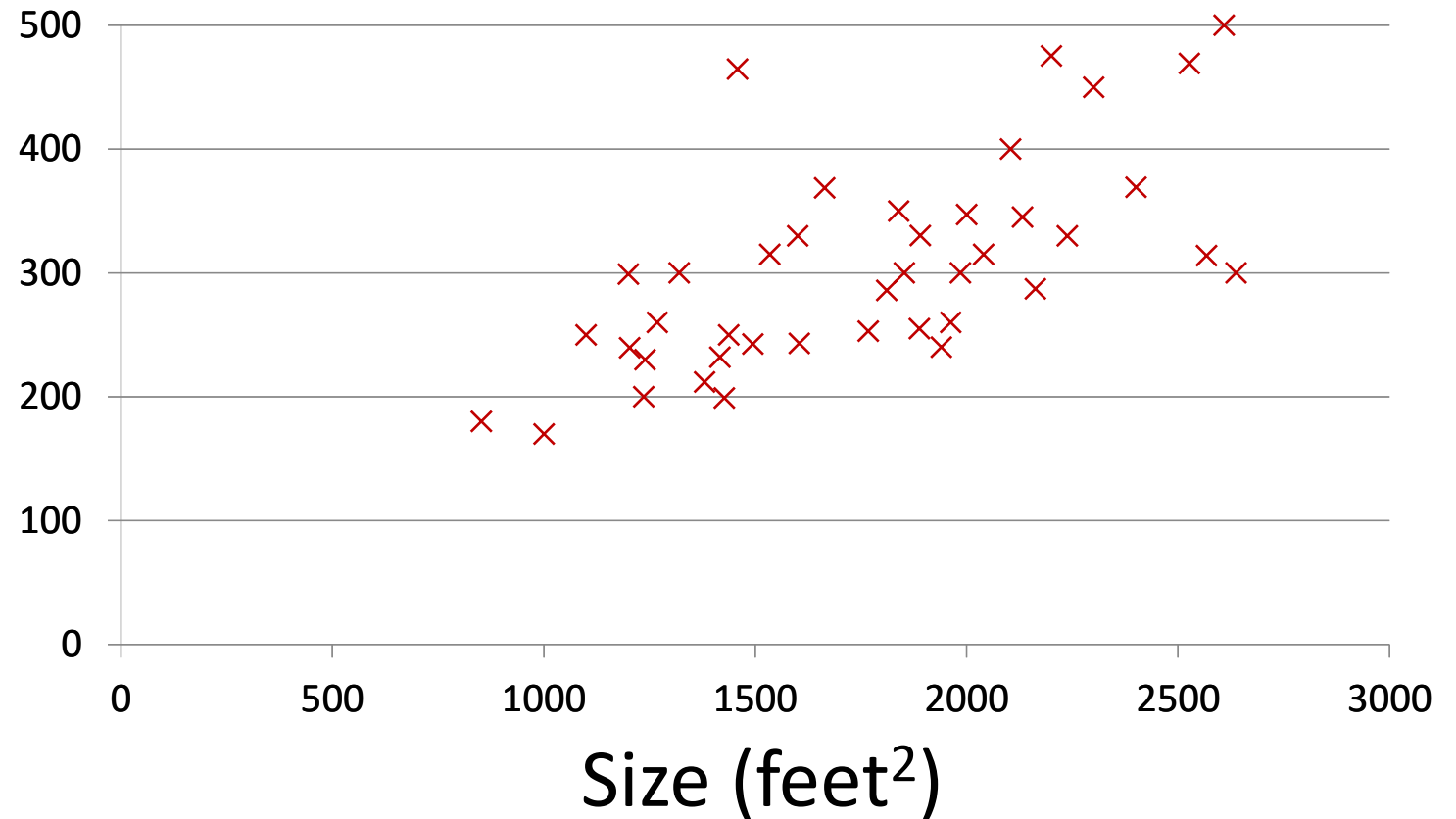
m = Number of training examples

x 's = “input” variable / features

y 's = “output” variable / “target” variable

Housing Prices (Portland, OR)

Price
(in 1000s
of dollars)



Supervised Learning

Given the “right answer” for each example in the data.

Regression Problem

Predict real-valued output



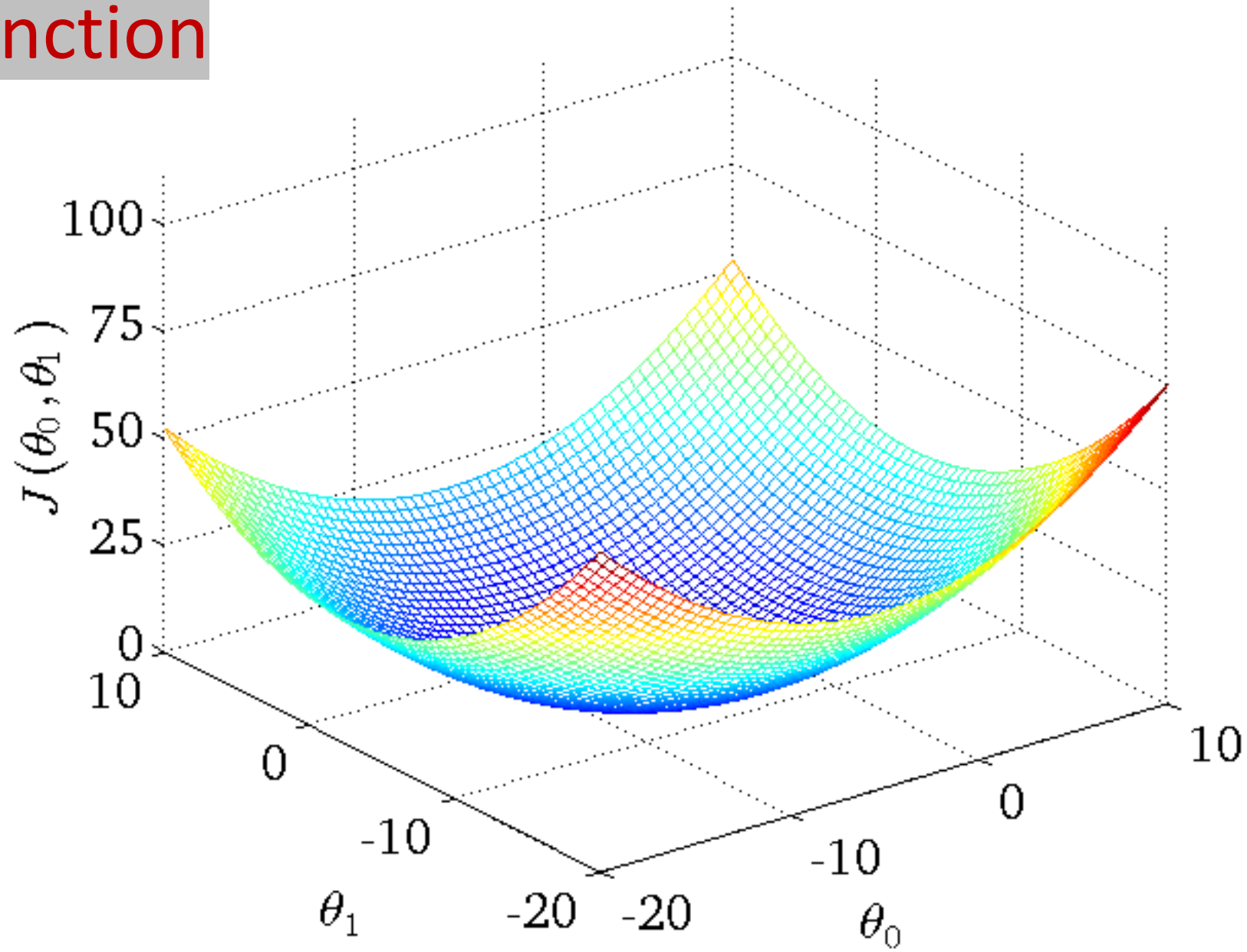
Hypothesis: $h_{\theta}(x) = \theta_0 + \theta_1 x$

Parameters: θ_0, θ_1

Cost Function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$

Goal: minimize $J(\theta_0, \theta_1)$
 θ_0, θ_1

Cost Function



Multiple features (variables).

Size (feet ²)	Price (\$1000)
x	y
2104	460
1416	232
1534	315
852	178
...	...

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Hypothesis: $h_{\theta}(x) = \theta^T x = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$

Parameters: $\theta_0, \theta_1, \dots, \theta_n$

Cost function:

$$J(\theta_0, \theta_1, \dots, \theta_n) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Gradient descent:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \dots, \theta_n)$$

} (simultaneously update for every $j = 0, \dots, n$)